MILITARY SPECIFICATION

ELEMENT, IGNITION PERCUSSION-ELECTRIC, M4 PARTS FOR

1. SCOPE

1.1 This specification covers the parts for one type of ignition element designated as the M4 Percussion-Electric Ignition Element.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition.
MIL-P-10025 - Packaging and Marking for Domestic Shipment of Inert Ammunition Components.
MIL-I-45208 - Inspection Requirements; General Specification for.
MIL-I-45607 - Inspection Equipment, Supply and Maintenance.

STANDARDS

FEDERAL

FED-STD-151 - Metal; Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.

FSC 1390

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DRAWINGS

ORDNANCE CORPS

7548455 - Insulator, Plunger.
7548456 - Washer, Plunger.
7548458 - Cup, Ignition.
7548459 - Ignition Cup Assembly.
7548460 - Pin, Plunger.
7548461 - Cup, Plunger.
7548462 - Plunger.
7548521 - Plunger Cup Assembly.

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
GL-7156104 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts.-The parts shall comply with all requirements specified on Drawings (dwgs.) 7548455, 7548456, 7548458, 7548459, 7548460, 7548461, 7548462, and 7548521 (as applicable), and with all requirements specified in applicable specifications.

3.3 Workmanship.-All parts shall be dry and free of burrs, chips, dirt, grease, rust, corrosion and other foreign material. Particular attention shall be given to freedom of the parts from burrs, sharp edges, cracks, etc. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-G-2550 shall apply.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of analysis on all material procured directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Preproduction inspection.

4.2.1 Submission.-The contractor shall submit 25 of each part and each sub-assembly under contract, which have been produced by the production process which the contractor will use in fulfilling the contract, to a Government-approved facility designated by the contracting officer. The preproduction quantity shall be submitted prior to initiation of production and whenever a change occurs in manufacturing process, material used, drawing, or specification, such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.-Each component and sub-assembly of the preproduction quantity shall be inspected for conformance to all drawing requirements (dimensional and otherwise) and with all the requirements of Section 3 of this specification.

4.2.3 Rejection.-If any component or sub-assembly fails to comply with the applicable requirements, the preproduction quantity shall be rejected and the contractor shall take the necessary corrective action and submit a new preproduction quantity, or portion thereof, as specified by the contracting officer. The contractor shall continue to submit new or supplemental quantities as necessary, until such time as a quantity successfully passes the inspection specified or until otherwise directed by the procuring activity.

4.3 Inspection provisions.

4.3.1 Lot formation.-A lot shall consist of parts and sub-assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot number.

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4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Pin, plunger (see dwg. 7548460).

Categories	Defects	Method of inspection
Critical:		
1.	Length	Gage
Major:	AQL 0.25 percent	
101.	Diameter, minimum (min.)	Gage
102.	Straightness, and diameter maximum (max.) (fails to pass freely through a gage of the dimensions specified on the applicable drawing)	Gage
Minor:	AQL 0.40 percent	
201.	Evidence of poor workmanship (see 3.3).	Visual

4.3.2.2 Washer, plunger (see dwg. 7548456).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.65 percent	
101.	Inside diameter	Gage
102.	Concentricity of outside diameter with inside diameter	Gage
103.	Thickness	Gage
104.	Deformation	Visual
105.	Burr or crack	Visual
106.	Corners not sharp	Visual
Minor:	AQL 1.50 percent	
201.	Outside diameter	Gage
202.	Finish incorrect	Visual
203.	Bare spot in protective coating exposing base metal	Visual
204.	Evidence of poor workmanship (see 3.3).	Visual

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4.3.2.3 Plunger (see dwg. 7548462).

Categories	Defects	Method of inspection
Critical:	None defined.	
Major:	AQL 1.00 percent	
101.	Length to flange	Gage
102.	Concentricity of larger diameter with flange	Gage
103.	Concentricity of smaller diameter with flange	Gage
104.	Diameter above flange	Gage
105.	Diameter below flange	Gage
106.	Diameter of plunger pin hole	Gage
107.	Chamfer missing from either end of plunger pin hole	Gage
108.	Profile of percussion point	Gage
109.	Location of plunger pin hole	Gage
110.	Length of spherical percussion point	Gage
111.	Diameter of flange	Gage
Minor:	AQL 2.50 percent	
201.	Thickness of flange	Gage
202.	Length from rear end to surface at base of percussion point	Gage
203.	Overall length, max.	Gage
204.	Radius at intersection of diameter rear of flange and rear surface of flange, max.	Visual
205.	Finish incorrect	Visual
206.	Evidence of poor workmanship (see 3.3).	Visual

4.3.2.4 Cup, ignition (see dwg. 7548458)

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Concentricity of outside diameter with large inside diameter	Gage
102.	Large inside diameter	Gage

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Minor:		AQL 2.50 percent
201.	Total length	Gage
202.	Small inside diameter	Gage
203.	Outside diameter	Gage
204.	Thickness through shoulder	Gage
205.	Chamfer missing (burr or sharp edge)	Visual
206.	Finish of interior surfaces incorrect	Visual
207.	Evidence of poor workmanship (see 3.3).	Visual

4.3.2.5 Cup, plunger (see dwg. 7548461).

Categories	Defects	Method of inspection
Critical:		
1.	Depth of plunger pin slot, max	Gage
2.	Depth to bottom of primer cavity, min.	Gage
Major:	AQL 0.40 percent	
101.	Concentricity of plunger cavity with outside diameter	Gage
102.	Concentricity of primer cavity with outside diameter	Gage
103.	Diameter of flash hole, min.	Gage
104.	Depth to bottom of plunger cavity	Gage
Minor:	AQL 4.00 percent	
201.	Total length	Gage
202.	Diameter of primer cavity	Gage
203.	Diameter of plunger cavity	Gage
204.	Diameter of flash hole, max.	Gage
205.	Width of plunger pin slot	Gage
206.	Depth of plunger pin slot, min.	Gage
207.	Depth to bottom of primer cavity, max.	Gage
208.	Location of plunger pin slot	Gage
209.	Outside diameter of slotted end	Gage
210.	Diameter of flash hole boss	Gage
211.	Length of flash hole boss	Gage
212.	Diameter of grooved surface	Gage
213.	Flash hole chamfer missing	Visual
214.	Annular grooves missing	Visual
215.	Finish in primer cavity incorrect	Visual
216.	Incorrect number of annular grooves	Visual
217.	Evidence of poor workmanship (see 3.3).	Visual

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4.3.2.6 Plunger cup assembly (see dwg. 7548521).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.65 percent	
101.	Concentricity of large outside diameter with large inside diameter	Gage
102.	Concentricity of small outside diameter with large inside diameter	Gage
103.	Large outside diameter of insulator, min.	Gage
104.	Small outside diameter of insulator, min.	Gage
105.	Insulator incompletely formed, cracked or broken	Visual
Minor:	AQL 4.00	
201.	Large outside diameter of insulator, max.	Gage
202.	Small outside diameter of insulator, max.	Gage
203.	Intermediate diameter of insulator	Gage
204.	Length from slotted end of plunger cup to insulator	Gage
205.	Length from boss end of plunger cup to large outside diameter of insulator.	Gage
206.	Length from boss end of plunger cup to intermediate diameter	Gage
207.	Length from boss end of plunger cup to insulator	Gage
208.	Evidence of poor workmanship (see 3.3).	Visual

4.3.2.7 Insulator, plunger (see dwg. 7548455).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
101.	Concentricity of small inside diameter with large inside diameter	Gage
102.	Small inside diameter	Gage
103.	Crack, split, perforation or similar defect	Visual
104.	Malformation	Visual

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Minor:		AQL 4.00 percent
201.	Small outside diameter	Gage
202.	Large inside diameter	Gage
203.	Large outside diameter	Gage
204.	Thickness of shoulder	Gage
205.	Length from rear end to shoulder	Gage
206.	Total length	Gage
207.	Excessive irregularities	Visual
208.	Evidence of poor workmanship (see 3.3).	Visual

4.3.2.8 Ignition cup assembly (see dwg. 7548459).

Categories	Defects	Method of inspection
Critical:	None defined.	
Major:	AQL 0.65 percent	
101.	Total length, max.	Gage
102.	Outside diameter of ignition cup, max..	Gage
103.	Bridge wire extends beyond outside diameter of ignition cup	Visual-Manual
104.	Component loose	Manaul
105.	Bridge wire broken	Visual
106.	Flash hole obstructed	Visual
Minor:	AQL 1.50 percent	
201.	Width of ignition cup slot, max.	Gage
202.	Depth of ignition cup slot	Gage
203.	Swaging of bridge wire missing or incomplete	Visual
204.	Evidence of poor workmanship (see 3.3).	Visual

4.3.3 Testing.

4.3.3.1 Rockwell hardness of plunger washer (see dwg. 7548456) Major defect.-The sampling plan for this test shall be in accordance with Standard MIL-STD-105, using code letter H and an AQL of 0.40 percent. The test shall be performed as specified in 4.4.1.

4.3.3.2 Electrical resistance (see dwg. 7548459) Major defect.-This test shall be conducted 100 percent. Any assembly which fails to comply with the resistance requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.2.

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4.3.4 Inspection equipment.-For the performance of the inspections listed herein, the contractor shall have available and utilize correctly all necessary gages, measuring and testing equipment identified by Index of Inspection Equipment Lists No. GL-7156104, or approved equivalents thereto. Designs of equipment developed by the contractor for use in lieu of that specified in the Index of Inspection Equipment Lists shall be documented and submitted to the Government for approval as equivalent (see 6.2). Approval shall be obtained prior to initiation of production. Supply and maintenance of inspection equipment shall be in accordance with Specification MIL-L-45607.

4.4 Test methods and procedures.

4.4.1 Rockwell hardness of plunger washer.-The plunger washer shall be tested as specified in Federal Standard 151, Method 243.

4.4.2 Electrical resistance.-The assembly shall be tested in accordance with the equipment Index (see 4.3.4).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (see 5.2).

5.2 Packing.

5.2.1 Level C.-All parts shall be packed for shipment in accordance with Specification MIL-P-10025. Ignition Cup Assemblies shall be packed in a manner to prevent damage to the Bridge Wire.

5.2.2 Marking.-Marking of all packing and packaging shall comply with Specification MIL-P-10025.

5.3 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

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6.2 Submission of contractor equipment designs.-Documentation of contractor developed equipment designs shall be submitted in duplicate to Picatinny Arsenal, Dover, New Jersey, ATTN: ORDBB-ND, for approval. This documentation shall identify all standard devices and set-ups etc., as well as special equipment designs.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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